Sustainable Design for a Self-Sufficient Tiny Home

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CHEE 204 - Water and Energy: Conventional and Alternate Systems

Introduction

Tiny homes have gained significant popularity in recent years due to their ability to promote a more minimalist lifestyle, reduce environmental impacts, and provide more affordable housing. Typically, these homes are under 400 square feet and create a sustainable living situation by reducing excessive space, materials, and energy in construction. With climate change concerns increasing on a global scale, natural resources being depleted, and urban sprawl becoming more prevalent, tiny homes present an opportunity for a sustainable, eco-friendly lifestyle. The design of a stationary, self-sufficient tiny home focuses on achieving energy independence and water sustainability through various integrated systems. By incorporating solar power, rainwater harvesting, greywater reuse, and gardening, the goal is to create a compact, yet fully functional living space that minimizes environmental impact. This approach will balance modern comfort and aesthetics with environmental consciousness, demonstrating how small spaces can seamlessly incorporate sustainable systems without sacrificing comfort, style, or functionality.

This 400 square foot home is designed as a permanent structure rather than one on wheels, allowing for greater integration of systems like solar panels, food gardens, and underground water storage. The house will be built using locally sourced materials such as reclaimed wood and recycled metal. An open floor plan maximizes natural lighting and airflow, minimizing the need for mechanical heating and cooling. Energy-efficient windows will account for 20% of the wall area, enhancing daylighting and reducing electricity demand. A green roof system covering 200 square feet will provide insulation and support food production and stormwater management, capable of retaining 1.5 inches of rainfall per event (Schade et al.,

2021). Built-in storage, convertible furniture, and multipurpose spaces reduce clutter while ensuring comfort and usability.

Energy and Roof Systems

To meet energy needs while optimizing limited space, the roof of the tiny home is designed as a multifunctional system that integrates photovoltaic panels with a lightweight green roof and rooftop garden modules. This layered approach makes use of vertical surface area while also offering environmental and thermal benefits.

The solar component will consist of a 3 kW photovoltaic system mounted on the south-facing section of the roof. In a high-sun region like Arizona, this system can produce approximately 14.5 kWh of electricity per day, roughly triple the estimated 3-4 kWh required by an energy-efficient 400 sq-ft tiny home (Tiny House Search, 2024). This surplus allows for reliable energy availability during cloudy days or periods of higher demand and will offset nearly 2.8 metric tons of CO_2 annually compared to relying on the grid. Electricity generated during the day will be stored in a 10 kWh lithium-ion battery bank, providing autonomy for 1-2 days without sunlight.. The home will include Energy Star appliances consuming at least 15% less energy than conventional models, LED lighting, and a mini-split heat pump. Passive solar design, including south-facing windows, thermal mass floors, and structural insulation, will help regulate temperature naturally and reduce HVAC demand.

The remaining rooftop area will support an extensive green roof and modular garden planters. The green roof contributes natural insulation and stormwater absorption, helping to regulate internal temperatures (Schade el al., 2021). Garden modules will grow shallow-rooted vegetables and herbs, using lightweight planting soil and drip irrigation sourced from captured

rainwater. Together, this hybrid roofing system maximizes space without compromising solar exposure or structural integrity.

Water Systems: Harvesting and Reuse Water

Water will be sourced through a rooftop rainwater harvesting system. With an average annual rainfall of 12 inches in Tucson and a 400 sq-ft roof, the home can collect up to 3,000 gallons of water per year (Pima County Public Works, n.d.).

$$400\ ft^2 \times\ 0.623\ gal/inch/ft^2 \times 12\ in\ =\ \sim 2990\ gallons/year$$

This supply, combined with a 1,500-gallon cistern and a gravity-fed filtration system, will cover most non-potable water needs, including irrigation and toilet flushing. A gravity-fed filtration system using sand and activated carbon will treat water for safe household use. According to the EPA, low-flow fixtures can reduce indoor water use by up to 30%, bringing daily consumption down to 25-30 gallons per person (EPA, 2000).

A waterless composting toilet will eliminate blackwater output and reduce overall household water demand by ~6,600 gallons/year per person (Edmonds, 2008). Greywater from sinks and showers, which is estimated to be 10-15 gallons per day, will be filtered and redirected to a drip irrigation system for the garden and surrounding landscape, closing the loop in water reuse (Drew, 2025; Ling, 2025).

Food Production: Permaculture in Small Spaces

The home includes integrated food systems that provide fresh produce and contribute to site resilience. A rooftop garden covering 100 square feet will grow herbs, leafy greens, and root vegetables in lightweight soil using drip irrigation. Each square foot can yield 0.5–1 lb of

produce monthly, providing an estimated 600–1,200 lbs of food annually from the combined rooftop and ground-level gardens (Vineyards, 2016).

On the ground, raised beds and vertical planters will support crops such as tomatoes, squash, and beans, while native pollinator plants enhance biodiversity and support yields. A rain garden located near the home will collect overflow from the rooftop system and serve as a filtration and wildlife habitat. Kitchen scraps will be composted in dual rotating bins, producing up to 200 pounds of nutrient-rich compost annually to support healthy soil and reduce landfill waste (Indiana University, 2020).

Conclusion

Designing a tiny home that prioritizes water and energy resilience requires thoughtful integration of systems that support independence without compromising design. Through a 3 kW solar system, a 10 kWh battery, and passive solar strategies, the home achieves year-round energy autonomy. Rainwater harvesting and greywater reuse provide nearly 3,000 gallons of water annually, supported by conservation plumbing and a composting toilet that saves over 6,000 gallons annually. Integrated permaculture yields hundreds of pounds of fresh produce each year.

This home model provides a compelling path toward living well within ecological limits, balancing environmental values with quality of life. As climate challenges intensify and housing costs rise, tiny homes like this one offer a replicable, data-backed model for sustainable living in arid urban environments.

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